

### 3.11.2 Thermodynamics and engines (A-level only)

#### 3.11.2.1 First law of thermodynamics (A-level only)

##### Content

Quantitative treatment of first law of thermodynamics,  $Q = \Delta U + W$

where  $Q$  is energy transferred to the system by heating,  $\Delta U$  is increase in internal energy and  $W$  is work done **by** the system.

Applications of first law of thermodynamics.

#### 3.11.2.2 Non-flow processes (A-level only)

##### Content

Isothermal, adiabatic, constant pressure and constant volume changes.

$$pV = nRT$$

adiabatic change :  $pV^\gamma = \text{constant}$

isothermal change :  $pV = \text{constant}$

at constant pressure  $W = p\Delta V$

Application of first law of thermodynamics to the above processes.

#### 3.11.2.3 The $p$ - $V$ diagram (A-level only)

##### Content

Representation of processes on  $p$ - $V$  diagram.

Estimation of work done in terms of area below the graph.

Extension to cyclic processes: *work done per cycle = area of loop*

Expressions for work done are not required except for the constant pressure case,  $W = p\Delta V$

Understanding of a four-stroke petrol engine cycle and a diesel engine cycle, and of the corresponding indicator diagrams.

Comparison with the theoretical diagrams for these cycles; use of indicator diagrams for predicting and measuring power and efficiency

$$\text{input power} = \text{calorific value} \times \text{fuel flow rate}$$

$$\text{Indicated power as } (\text{area of } p-V \text{ loop}) \times (\text{no. of cycles per second}) \times (\text{no. of cylinders})$$

$$\text{Output or brake power, } P = T\omega$$

$$\text{friction power} = \text{indicated power} - \text{brake power}$$

Engine efficiency; overall, thermal and mechanical efficiencies.

$$\text{Overall efficiency} = \frac{\text{brake power}}{\text{input power}}$$

$$\text{Thermal efficiency} = \frac{\text{indicated power}}{\text{input power}}$$

$$\text{Mechanical efficiency} = \frac{\text{brake power}}{\text{indicated power}}$$

A knowledge of engine constructional details is not required.

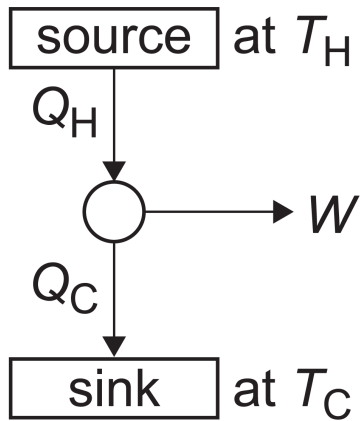
Questions may be set on other cycles, but they will be interpretative and all essential information will be given.

Impossibility of an engine working only by the First Law.

Second Law of Thermodynamics expressed as the need for a heat engine to operate between a source and a sink.

$$\text{efficiency} = \frac{W}{Q_H} = \frac{Q_H - Q_C}{Q_H}$$

$$\text{maximum theoretical efficiency} = \frac{T_H - T_C}{T_H}$$

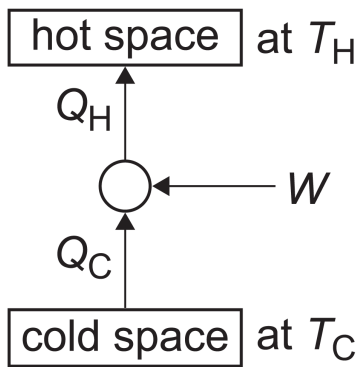


Reasons for the lower efficiencies of practical engines.

Maximising use of  $W$  and  $Q_H$  for example in combined heat and power schemes.

Basic principles and uses of heat pumps and refrigerators.

A knowledge of practical heat pumps or refrigerator cycles and devices is not required.



Coefficients of performance:

$$\text{refrigerator: } COP_{\text{ref}} = \frac{Q_C}{W} = \frac{Q_C}{Q_H - Q_C} = \frac{T_C}{T_H - T_C}$$

$$\text{heat pump: } COP_{\text{hp}} = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_C} = \frac{T_H}{T_H - T_C}$$